

Momentum-Space QCD Sum-Rule Setup for the $B_c(1P)$ Axial-Vector Mixing Angle

1 Purpose and conventions

We study the mixing between the two axial-vector $B_c(1P)$ basis states associated with the 3P_1 and 1P_1 structures. The current implementation follows the Aliev et al. mixing-angle method, but the OPE is evaluated in momentum space because both quark lines are heavy.

This note records the calculation implemented in `BcMixingMomentum.wl`. It is written so that every convention and intermediate object can be checked independently before using the numerical angle in a paper.

The quark content convention is

$$B_c^+ = \bar{b}c. \quad (1)$$

The two interpolating currents are taken as

$$J_\mu^A = \bar{b} \gamma_\mu \gamma_5 c, \quad (2)$$

$$J_\mu^B = i \bar{b} \sigma_{\mu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 c, \quad (3)$$

where

$$\sigma_{\mu\nu} = \frac{i}{2} [\gamma_\mu, \gamma_\nu]. \quad (4)$$

The explicit factor of i in J_μ^B is retained. This is important because `FeynCalc`'s `DiracSigma` already contains the conventional $i/2$ commutator.

The factor $1/(m_b + m_c)$ is also essential. Since $[\bar{Q}Q] = 3$ and $[p^\alpha] = 1$, the unnormalized tensor current $i\bar{b}\sigma_{\mu\alpha}p^\alpha\gamma_5c$ has mass dimension 4, while J_μ^A has mass dimension 3. The normalized current above makes both currents dimension 3.

If the opposite flavor convention is used, for example $B_c^- = \bar{c}b$, the sign of the quoted angle can change. The absolute convention must therefore be kept fixed when comparing to other papers.

2 Mass-dimension check

The quark field has dimension $3/2$, while p^α , m_b , and m_c have dimension 1. Therefore

$$[J_\mu^A] = 3, \quad [J_\mu^B] = 3 \quad (5)$$

only after the tensor current is normalized by $m_b + m_c$.

For currents with dimensions d_i and d_j ,

$$[\Pi_{\mu\nu}^{ij}] = d_i + d_j - 4, \quad (6)$$

because of the d^4x integration in the two-point function. Thus the normalized basis gives

$$[\Pi_1^{AA}] = [\Pi_1^{AB}] = [\Pi_1^{BB}] = 2. \quad (7)$$

The spectral densities have the same mass dimension as the invariant correlator,

$$[\rho^{AA}] = [\rho^{AB}] = [\rho^{BB}] = 2. \quad (8)$$

In the present code the Borel moment is evaluated as $\int ds e^{-s/M^2} \rho(s)$, so all three numerical moments carry one common extra factor of $[ds] = 2$. This common convention does not affect the mixing angle.

If the tensor current were left unnormalized, the dimensions would instead be

$$[\rho^{AA}], [\rho^{AB}], [\rho^{BB}] = 2, 3, 4, \quad (9)$$

and the matrix entries in the angle formula would not be dimensionally compatible.

3 Implementation map

The main code objects are:

- `Correlator[channel, order]`: $\Pi_1^{ij}(p^2)$ after FeynCalc Dirac algebra and spin projection.
- `SpectralDensity[channel, ‘‘pert’’, s]`: $\rho_{\text{pert}}^{ij}(s)$.
- `NumericBorelPi[channel, order, M2, s0]`: $\Pi_{\text{order}}^{ij}(M^2, s_0)$.
- `NumericMixingAngleDegrees[M2, s0, ‘‘total’’]`: $\theta(M^2, s_0)$ from $\Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij} + \Pi_{G^3}^{ij}$.

The implemented OPE orders are

$$\text{pert}, \quad G^2, \quad \text{pertG2} = \text{pert} + G^2, \quad G^3, \quad \text{total} = \text{pert} + G^2 + G^3. \quad (10)$$

The implemented dimension-6 triple-gluon condensate $\langle g_s^3 G^3 \rangle$ is the standard vacuum-averaged single-line propagator term. Possible cross-line open-field G^3 terms need a separate derivation.

4 Correlators and spin-1 projection

Define

$$\Pi_{\mu\nu}^{ij}(p) = i \int d^4x e^{ip \cdot x} \langle 0 | T \{ J_\mu^i(x) J_\nu^{j\dagger}(0) \} | 0 \rangle, \quad i, j \in \{A, B\}. \quad (11)$$

The invariant amplitude used for the axial-vector spin-1 state is extracted with

$$P_{(1)}^{\mu\nu} = \frac{1}{3} \left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right), \quad (12)$$

so that

$$\Pi_1^{ij}(p^2) = P_{(1)}^{\mu\nu} \Pi_{\mu\nu}^{ij}(p). \quad (13)$$

The mixing angle is then obtained from

$$\tan 2\theta = -\frac{2\Pi_1^{AB}}{\Pi_1^{AA} - \Pi_1^{BB}}. \quad (14)$$

Numerically we use the quadrant-safe definition

$$\theta_{\text{raw}} = \frac{1}{2} \text{atan2}(-2\Pi_1^{AB}, \Pi_1^{AA} - \Pi_1^{BB}), \quad (15)$$

and quote the principal branch

$$\theta = \theta_{\text{raw}} - \frac{\pi}{2} \text{Round} \left(\frac{\theta_{\text{raw}}}{\pi/2} \right), \quad \theta \in [-45^\circ, 45^\circ]. \quad (16)$$

5 Momentum-space OPE

The perturbative heavy-quark propagator is

$$S_Q^{(0)}(k) = \frac{\not{k} + m_Q}{k^2 - m_Q^2}. \quad (17)$$

For the dimension-4 gluon condensate we use the background-field expansion

$$S_Q(k) = S_Q^{(0)}(k) + g_s G_A^{\alpha\beta} T^A S_{Q,\alpha\beta}^{(G)}(k) + S_Q^{(G^2)}(k), \quad (18)$$

where

$$S_{Q,\alpha\beta}^{(G)}(k) = -\frac{1}{4} \frac{\sigma_{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma_{\alpha\beta}}{(k^2 - m_Q^2)^2}, \quad (19)$$

$$S_Q^{(G^2)}(k) = \frac{\langle g_s^2 G^2 \rangle}{12} m_Q \frac{k^2 + m_Q}{(k^2 - m_Q^2)^4}. \quad (20)$$

Equivalently,

$$g_s G_A^{\alpha\beta} T^A S_{Q,\alpha\beta}^{(G)}(k) = -\frac{g_s G_A^{\alpha\beta} T^A}{4} \frac{\sigma_{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma_{\alpha\beta}}{(k^2 - m_Q^2)^2}. \quad (21)$$

The one-gluon fields are contracted with

$$\langle g_s^2 G_A^{\alpha\beta} G_B^{\rho\sigma} \rangle = \frac{\delta^{AB}}{96} \langle g_s^2 G^2 \rangle \left(g^{\alpha\rho} g^{\beta\sigma} - g^{\alpha\sigma} g^{\beta\rho} \right), \quad (22)$$

and

$$\text{Tr}(T^A T^B) = \frac{\delta^{AB}}{2}. \quad (23)$$

Thus the color factor for the $S^{(G)}S^{(G)}$ piece is

$$\frac{N_c^2 - 1}{192} \langle g_s^2 G^2 \rangle, \quad N_c = 3. \quad (24)$$

The three projected OPE contributions are generated from

$$\Pi_1^{AA} \sim P_{(1)}^{\mu\nu} \text{Tr} [\gamma_\mu \gamma_5 S_c(k) \gamma_\nu \gamma_5 S_b(k-p)], \quad (25)$$

$$\Pi_1^{AB} \sim P_{(1)}^{\mu\nu} \text{Tr} \left[\gamma_\mu \gamma_5 S_c(k) i\sigma_{\nu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 S_b(k-p) \right], \quad (26)$$

$$\Pi_1^{BB} \sim P_{(1)}^{\mu\nu} \text{Tr} \left[i\sigma_{\mu\alpha} \frac{p^\alpha}{m_b + m_c} \gamma_5 S_c(k) i\sigma_{\nu\beta} \frac{p^\beta}{m_b + m_c} \gamma_5 S_b(k-p) \right]. \quad (27)$$

6 Perturbative spectral densities

For the perturbative loop, impose the two-particle cut

$$k^2 = m_c^2, \quad (k-p)^2 = m_b^2, \quad (28)$$

which gives

$$k \cdot p = \frac{s + m_c^2 - m_b^2}{2}, \quad s = p^2. \quad (29)$$

Define

$$\lambda(s, m_b, m_c) = s^2 + m_b^4 + m_c^4 - 2sm_b^2 - 2sm_c^2 - 2m_b^2 m_c^2. \quad (30)$$

The perturbative spectral densities are implemented as

$$\rho_{\text{pert}}^{ij}(s) = \frac{N_c}{16\pi^2} \frac{\sqrt{\lambda(s, m_b, m_c)}}{s} \frac{N^{ij}(s)}{(m_b + m_c)^{n_B^{ij}}}, \quad (31)$$

where $n_B^{AA} = 0$, $n_B^{AB} = n_B^{BA} = 1$, and $n_B^{BB} = 2$ counts how many tensor currents occur in the channel. The unnormalized numerator functions are

$$N^{AA}(s) = -\frac{2}{s} [(m_b + m_c)^2 - s] [(m_b - m_c)^2 + 2s], \quad (32)$$

$$N^{AB}(s) = 6(m_b - m_c) [(m_b + m_c)^2 - s], \quad (33)$$

$$N^{BB}(s) = 2 [-2(m_b^2 - m_c^2)^2 + (m_b^2 - 6m_b m_c + m_c^2)s + s^2]. \quad (34)$$

Equivalently, after setting $N_c = 3$,

$$\rho_{\text{pert}}^{AA}(s) = -\frac{3}{8\pi^2 s^2} [(m_b + m_c)^2 - s] [(m_b - m_c)^2 + 2s] \sqrt{\lambda(s, m_b, m_c)}, \quad (35)$$

$$\rho_{\text{pert}}^{AB}(s) = \frac{9}{8\pi^2 s(m_b + m_c)} (m_b - m_c) [(m_b + m_c)^2 - s] \sqrt{\lambda(s, m_b, m_c)}, \quad (36)$$

$$\rho_{\text{pert}}^{BB}(s) = \frac{3}{8\pi^2 s(m_b + m_c)^2} [-2(m_b^2 - m_c^2)^2 + (m_b^2 - 6m_b m_c + m_c^2)s + s^2] \sqrt{\lambda(s, m_b, m_c)}. \quad (37)$$

The perturbative Borel moments are

$$\Pi_{\text{pert}}^{ij}(M^2, s_0) = \int_{(m_b + m_c)^2}^{s_0} ds e^{-s/M^2} \rho_{\text{pert}}^{ij}(s). \quad (38)$$

With this convention all three perturbative spectral densities have mass dimension 2. Without the factors of $(m_b + m_c)^{-n_B^{ij}}$, their dimensions would be 2, 3, and 4 for AA , AB , and BB , respectively.

7 Dimension-4 and dimension-6 gluon condensates

The implemented condensate contribution is

$$\Pi_{G^2}^{ij} = \Pi^{ij}[S_c^{(G^2)} S_b^{(0)}] + \Pi^{ij}[S_c^{(0)} S_b^{(G^2)}] + \Pi^{ij}[S_c^{(G)} S_b^{(G)}]. \quad (39)$$

After Feynman parametrization the terms have the schematic form

$$\mathcal{A}_{G^2}^{ij}(x_1, x_2, s) = \sum_r C_r^{ij}(x_1, x_2, s) [m_c^2 x_1^2 + (m_b^2 + m_c^2 - s)x_1 x_2 + m_b^2 x_2^2]^{-n_r}. \quad (40)$$

The code then fixes the simplex by

$$x_1 = x, \quad x_2 = 1 - x. \quad (41)$$

This gives

$$\Delta(x, s) = m_c^2 x + m_b^2 (1 - x) - s x (1 - x). \quad (42)$$

For the triple-gluon condensate we use the standard momentum-space vacuum-averaged heavy-quark propagator term

$$S_Q^{(G^3)}(k) = \frac{\langle g_s^3 G^3 \rangle}{48} \frac{(\not{k} + m_Q) [\not{k}(k^2 - 3m_Q^2) + 2m_Q(2k^2 - m_Q^2)] (\not{k} + m_Q)}{(k^2 - m_Q^2)^6}. \quad (43)$$

The implemented dimension-6 contribution is

$$\Pi_{G^3}^{ij} = \Pi^{ij}[S_c^{(G^3)} S_b^{(0)}] + \Pi^{ij}[S_c^{(0)} S_b^{(G^3)}]. \quad (44)$$

This is the standard single-heavy-propagator contribution. It does not yet include any possible cross-line open-field terms of the schematic type $S^{(G_{\text{open}}^2)} S^{(G)}$, which require an explicitly open two-gluon propagator and a separate color/Lorentz derivation.

The G^2 and G^3 terms are not treated as ordinary smooth spectral densities. We instead use a direct Borel transform. Set $s = -Q^2$ and write

$$\Delta(x, -Q^2) = x(1-x) [Q^2 + \bar{s}(x)], \quad (45)$$

where

$$\bar{s}(x) = \frac{m_c^2 x + m_b^2 (1-x)}{x(1-x)}. \quad (46)$$

The Feynman-parameter amplitude is decomposed as

$$-i \mathcal{A}_{Gr}^{ij}(x, -Q^2) = \sum_{\ell \geq 1} \frac{a_\ell^{ij}(x)}{[Q^2 + \bar{s}(x)]^\ell} + \text{polynomial subtraction terms}. \quad (47)$$

The polynomial subtraction terms vanish after the Borel transform. The working phase convention is the explicit factor $-i$ shown above. This is a bookkeeping convention for removing the loop-integration prefactor returned by FeynCalc's Feynman-parameter amplitude. This is the convention that should be independently checked.

We use

$$\mathcal{B}_{M^2} \left[\frac{1}{(Q^2 + \bar{s})^n} \right] = \frac{1}{(n-1)! (M^2)^{n-1}} e^{-\bar{s}/M^2}. \quad (48)$$

Thus

$$\Pi_{Gr}^{ij}(M^2, s_0) = \int_{x_-}^{x_+} dx e^{-\bar{s}(x)/M^2} \sum_{\ell \geq 1} \frac{a_\ell^{ij}(x)}{(\ell-1)! (M^2)^{\ell-1}}. \quad (49)$$

The continuum restriction is

$$\bar{s}(x) \leq s_0, \quad (50)$$

which gives

$$x_\pm = \frac{s_0 + m_b^2 - m_c^2 \pm \sqrt{\lambda(s_0, m_b, m_c)}}{2s_0}. \quad (51)$$

8 Total Borel moments and numerical test point

The total moments used in the current numerical result are

$$\Pi^{ij}(M^2, s_0) = \Pi_{\text{pert}}^{ij}(M^2, s_0) + \Pi_{G^2}^{ij}(M^2, s_0) + \Pi_{G^3}^{ij}(M^2, s_0). \quad (52)$$

The current default numerical inputs are

$$m_b = 4.18 \text{ GeV}, \quad (53)$$

$$m_c = 1.27 \text{ GeV}, \quad (54)$$

$$\langle g_s^2 G^2 \rangle = 4\pi^2 (0.012) \text{ GeV}^4, \quad (55)$$

$$\langle g_s^3 G^3 \rangle = 0.57 \text{ GeV}^6, \quad (56)$$

$$M^2 = 10 \text{ GeV}^2, \quad (57)$$

$$s_0 = 55 \text{ GeV}^2. \quad (58)$$

For this point,

$$x_- = 0.3322728664, \quad x_+ = 0.9560816791. \quad (59)$$

The perturbative moments are

$$\Pi_{\text{pert}}^{AA} = 0.1441716826, \quad (60)$$

$$\Pi_{\text{pert}}^{AB} = -0.1051577192, \quad (61)$$

$$\Pi_{\text{pert}}^{BB} = 0.1348134513. \quad (62)$$

The gluon-condensate moments are

$$\Pi_{G^2}^{AA} = -0.0003658277, \quad (63)$$

$$\Pi_{G^2}^{AB} = 0.0002309420, \quad (64)$$

$$\Pi_{G^2}^{BB} = -0.0001888869. \quad (65)$$

The triple-gluon-condensate moments are

$$\Pi_{G^3}^{AA} = 0.0000700979, \quad (66)$$

$$\Pi_{G^3}^{AB} = -0.0000493659, \quad (67)$$

$$\Pi_{G^3}^{BB} = 0.0000310238. \quad (68)$$

Thus

$$\Pi_{\text{total}}^{AA} = 0.1438759528, \quad (69)$$

$$\Pi_{\text{total}}^{AB} = -0.1049761432, \quad (70)$$

$$\Pi_{\text{total}}^{BB} = 0.1346555882. \quad (71)$$

The corresponding angles are

$$\theta_{\text{pert}} = 43.726118943^\circ, \quad (72)$$

$$\theta_{\text{pert}+G^2} = 43.747426660^\circ, \quad (73)$$

$$\theta_{\text{pert}+G^2+G^3} = 43.742693524^\circ. \quad (74)$$

At this point the relative condensate corrections are

$$\frac{\Pi_{G^2}^{AA}}{\Pi_{\text{pert}}^{AA}} = -0.002537, \quad (75)$$

$$\frac{\Pi_{G^2}^{AB}}{\Pi_{\text{pert}}^{AB}} = -0.002196, \quad (76)$$

$$\frac{\Pi_{G^2}^{BB}}{\Pi_{\text{pert}}^{BB}} = -0.001401, \quad (77)$$

$$\frac{\Pi_{G^3}^{AA}}{\Pi_{\text{pert}}^{AA}} = 0.000486, \quad (78)$$

$$\frac{\Pi_{G^3}^{AB}}{\Pi_{\text{pert}}^{AB}} = 0.000469, \quad (79)$$

$$\frac{\Pi_{G^3}^{BB}}{\Pi_{\text{pert}}^{BB}} = 0.000230. \quad (80)$$

9 Perturbative α_s sensitivity model

The full perturbative $O(\alpha_s)$ correction is not yet calculated. As a diagnostic, the code implements a channel-dependent trial rescaling of the leading perturbative moment,

$$\Pi_{\text{pert}}^{ij}(M^2, s_0) \longrightarrow \left[1 + \frac{\alpha_s}{\pi} K_{ij}\right] \Pi_{\text{pert}}^{ij}(M^2, s_0). \quad (81)$$

The condensate pieces are left unchanged, so for the current total moment

$$\Pi_K^{ij} = \left[1 + \frac{\alpha_s}{\pi} K_{ij}\right] \Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij} + \Pi_{G^3}^{ij}. \quad (82)$$

This is only a sensitivity model; it is not a substitute for the real two-loop NLO perturbative spectral density.

For example, with $\alpha_s = 0.26$ and

$$K_{AA} = 1, \quad K_{AB} = 0, \quad K_{BB} = -1, \quad (83)$$

the total angle at $M^2 = 10 \text{ GeV}^2$, $s_0 = 55 \text{ GeV}^2$ changes from

$$\theta_{\text{base}} = 43.742693524^\circ \quad (84)$$

to

$$\theta_K = 40.625721673^\circ, \quad \Delta\theta = -3.116971851^\circ. \quad (85)$$

A coarse scan with $K_{AA}, K_{AB}, K_{BB} \in \{-2, 0, 2\}$ gives a maximum branch-corrected shift of about

$$\max |\Delta\theta| \simeq 7.52^\circ. \quad (86)$$

10 Monte Carlo uncertainty propagation

The numerical input table gives central values,

$$m_b = 4.18 \text{ GeV}, \quad m_c = 1.27 \text{ GeV}, \quad (87)$$

$$\langle g_s^2 G^2 \rangle = 4\pi^2 (0.012) \text{ GeV}^4, \quad \langle g_s^3 G^3 \rangle = 0.57 \text{ GeV}^6. \quad (88)$$

For the final quoted error these inputs should be assigned allowed intervals or probability distributions. In the present implementation we use uniform sampling over editable intervals. A Monte Carlo point is

$$X_i = \left\{ m_b^{(i)}, m_c^{(i)}, G_2^{(i)}, G_3^{(i)}, M_i^2, s_{0,i} \right\}, \quad (89)$$

where

$$G_2 \equiv \langle g_s^2 G^2 \rangle, \quad G_3 \equiv \langle g_s^3 G^3 \rangle. \quad (90)$$

Each component is drawn from the user-specified interval

$$x^{(i)} \sim U(x_{\min}, x_{\max}). \quad (91)$$

Fixed parameters are implemented by setting $x_{\min} = x_{\max}$, or in the Mathematica code by passing a single number instead of an interval.

The current working ranges in the code are

$$m_b \in [4.16, 4.21] \text{ GeV}, \quad m_c \in [1.25, 1.29] \text{ GeV}, \quad (92)$$

$$G_2 \in 4\pi^2 [0.006, 0.018] \text{ GeV}^4, \quad G_3 \in [0.28, 0.86] \text{ GeV}^6, \quad (93)$$

$$M^2 \in [8, 12] \text{ GeV}^2, \quad s_0 \in [50, 60] \text{ GeV}^2. \quad (94)$$

Equivalently, in the central-value-plus-error notation useful for the paper,

$$m_b = (4.185 \pm 0.025) \text{ GeV}, \quad m_c = (1.27 \pm 0.02) \text{ GeV}, \quad (95)$$

$$G_2 = 4\pi^2(0.012 \pm 0.006) \text{ GeV}^4, \quad G_3 = (0.57 \pm 0.29) \text{ GeV}^6, \quad (96)$$

$$M^2 = (10 \pm 2) \text{ GeV}^2, \quad s_0 = (55 \pm 5) \text{ GeV}^2. \quad (97)$$

These are not final physics choices. They are placeholders for the scan layer and should be replaced by the intervals adopted in the analysis.

For every random point we impose the physical continuum condition

$$s_{0,i} > \left(m_b^{(i)} + m_c^{(i)} \right)^2. \quad (98)$$

Accepted points are evaluated with the same OPE formula as above,

$$\theta_i = \frac{1}{2} \text{ArcTan} \left[\Pi_1^{AA}(X_i) - \Pi_1^{BB}(X_i), -2\Pi_1^{AB}(X_i) \right], \quad (99)$$

using the chosen OPE order, normally “total” for $\Pi_{\text{pert}} + \Pi_{G^2} + \Pi_{G^3}$. For the present fast uncertainty scan we keep the triple-gluon term switched off by default. In the Mathematica implementation this is controlled by `$BcMixingMonteCarloIncludeG3=False` or by the option `"IncludeG3" -> False`. With this switch, a requested “total” Monte Carlo scan is evaluated as “pertG2”,

$$\Pi_{\text{MC}}^{ij} = \Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij}, \quad (100)$$

and the G^3 contribution can be restored later by setting `"IncludeG3" -> True`.

The central value and Gaussian-width estimate are then

$$\bar{\theta} = \frac{1}{N} \sum_{i=1}^N \theta_i, \quad \sigma_{\theta, \text{Gauss}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\theta_i - \bar{\theta})^2}. \quad (101)$$

The code also reports the unbiased sample standard deviation, median, minimum, maximum, and the 16% and 84% quantiles. These quantiles are useful because the resulting θ distribution may not be exactly Gaussian.

The Mathematica entry point is

```
nSamples = 500;
histBins = 25;
m2Range = {8.0, 12.0};
s0Range = {50.0, 60.0};

ranges = Join[
  $BcMixingDefaultUncertaintyRanges,
  <|"M2" -> m2Range, "s0" -> s0Range|>
];

uncertaintyRun = MonteCarloMixingAngleUncertainty[
  nSamples, ranges, "total",
  "IncludeG3" -> False,
  "Seed" -> 1234,
  "Progress" -> True
];
uncertaintyRun["Summary"]
MonteCarloMixingAngleHistogram[uncertaintyRun, histBins]
MonteCarloMixingAnglePublicationHistogram[uncertaintyRun, histBins]
```


where `ranges` is an association such as

```
<|
  "mb" -> {4.16, 4.21},
  "mc" -> {1.25, 1.29},
  "G2" -> {4 Pi^2 0.006, 4 Pi^2 0.018},
  "G3" -> {0.28, 0.86},
  "M2" -> {8.0, 12.0},
  "s0" -> {50.0, 60.0}
|>
```

The number of random points, the input intervals, and the Borel window are therefore all editable without touching the algebra.

The first argument, for example 500, is the number of random accepted Monte Carlo points. The option `"Seed" -> 1234` fixes the random-number seed so the sample is reproducible. The Monte Carlo result is called `uncertaintyRun`, not `mc`, to avoid confusion with the charm mass parameter `"mc"`. To vary only the Borel window and the continuum threshold while keeping the QCD inputs fixed at their central values, one can use

```
ranges = Join[
  $BcMixingDefaultUncertaintyRanges,
  <|
    "mb" -> 4.18,
    "mc" -> 1.27,
    "G2" -> 4 Pi^2 0.012,
    "G3" -> 0.57,
    "M2" -> {8.0, 12.0},
    "s0" -> {50.0, 60.0}
  |>
];
```

The accepted random points and the summary can be exported for replotting:

```
ExportMonteCarloMixingAngleSamples[uncertaintyRun, "BcMixingMonteCarloSamples.csv"];
ExportMonteCarloMixingAngleSummary[uncertaintyRun, "BcMixingMonteCarloSummary.csv"];
Export["BcMixingMonteCarloHistogram.pdf",
  MonteCarloMixingAnglePublicationHistogram[uncertaintyRun, 25]];
```

11 How to reproduce the numerical check

In Mathematica, load the implementation with

```
SetDirectory["/Users/sbilmis/Bc_mixing"];
Get["BcMixingMomentum.wl"];
```

Then evaluate

```
NumericOPESummary[10, 55]
NumericMixingAngleDegrees[10, 55, "pert"]
NumericMixingAngleDegrees[10, 55, "pertG2"]
NumericMixingAngleDegrees[10, 55, "total"]
KFactorSensitivityRecord[10, 55, KFactorAssociation[1, 0, -1], 0.26, "total"]
KFactorSensitivityEnvelope[10, 55, {-2, 2, 2}, 0.26, "total"]
MixingMassDimensionReport[]
CheckMixingMassDimensions[]
PerturbativeSpectralDensityDimensionReport[]
```

For window studies one can scan

```
MixingAngleTable[{8, 12, 1}, {50, 60, 5}, "total"]
OPEConvergenceDataset[{8, 12, 1}, {50, 60, 5}]
uncertaintyRun = MonteCarloMixingAngleUncertainty[50, $BcMixingDefaultUncertaintyRanges,
  "total", "IncludeG3" -> False, "Seed" -> 1234]
uncertaintyRun["Summary"]
```

12 Auxiliary-parameter stability plots

As a comparison with the vector B_c^* analysis of Wang, we also consider the auxiliary window

$$M^2 \in [7.0, 9.0] \text{ GeV}^2, \quad s_0 = (54 \pm 1) \text{ GeV}^2. \quad (102)$$

The Mathematica implementation stores this as

```
$BcMixingWangWindow
```

and produces publication-style stability plots with

```
stabilityOrder = "pertG2";
```

```
m2Stability = MixingAngleStabilityM2PublicationPlot[
  $BcMixingWangWindow["M2Range"],
  $BcMixingWangWindow["s0Values"],
  stabilityOrder,
  $BcMixingDefaultParameters,
  "NPoints" -> 25,
  "YHalfWidth" -> 1.0
];
```

```
s0Stability = MixingAngleStabilityS0PublicationPlot[
  $BcMixingWangWindow["s0Range"],
  $BcMixingWangWindow["M2Values"],
  stabilityOrder,
  $BcMixingDefaultParameters,
  "NPoints" -> 25,
  "YHalfWidth" -> 1.0
];
```

The option "YHalfWidth" -> 1.0 prevents the vertical axis from being over-zoomed; it enforces at least a $\pm 1^\circ$ display window around the calculated angles. This is useful when the goal is to demonstrate stability with respect to the auxiliary parameters, not to magnify tiny numerical variations. For publication labels the plotting functions try to use the Mathematica package MaTeX automatically,

```
"UseMaTeX" -> Automatic
```

so that the axes and legends are rendered as $M^2(\text{GeV}^2)$, $s_0(\text{GeV}^2)$, θ° , and entries such as $s_0 = 54 \text{ GeV}^2$. If MaTeX is not installed, the code falls back to ordinary Mathematica labels with a degree symbol. The fallback can be forced with "UseMaTeX" -> False.

The plots and their underlying data can be exported with

```

Export["BcMixingThetaVsM2_WangWindow.pdf", m2Stability["Plot"]];
Export["BcMixingThetaVsS0_WangWindow.pdf", s0Stability["Plot"]];
ExportStabilityDataCSV[m2Stability["Data"],
  "BcMixingThetaVsM2_WangWindow.csv", "M2"];
ExportStabilityDataCSV[s0Stability["Data"],
  "BcMixingThetaVsS0_WangWindow.csv", "S0"];

```

13 Checklist for independent verification

A theorist can check the calculation in the following order:

1. Confirm the current convention, especially the explicit factor of i in J_μ^B , the normalization $p^\alpha/(m_b + m_c)$, and the definition of $\sigma_{\mu\nu}$.
2. Reproduce the three projected perturbative traces and verify the numerator functions $N^{AA}(s)$, $N^{AB}(s)$, and $N^{BB}(s)$.
3. Check the two-body phase-space normalization $N_c\sqrt{\lambda}/(16\pi^2s)$ used in $\rho_{\text{pert}}^{ij}(s)$.
4. Verify the background-field propagator terms and the color factor in the $S^{(G)}S^{(G)}$ contribution.
5. Independently derive the direct Borel formula for the G^2 and G^3 terms, including the $-i$ phase convention.
6. Check whether the standard single-line G^3 propagator contribution is sufficient, or whether cross-line open-field G^3 terms must be added.
7. Check that $\Pi_{G^2}/\Pi_{\text{pert}}$ and $\Pi_{G^3}/\Pi_{\text{pert}}$ remain small in the accepted Borel window.
8. Use the K_{ij} sensitivity model only as an estimate of possible channel-dependent perturbative $O(\alpha_s)$ uncertainty.
9. Derive the true $O(\alpha_s)$ two-loop perturbative spectral densities separately.
10. Determine the final working window using OPE convergence, Borel stability, and pole-dominance criteria.